BRIEF COMMUNICATIONS

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Non-Boussinesq effect: Asymmetric velocity profiles in thermal convection

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In thermal convection at high Rayleigh numbers, in the hard turbulent regime, a large scale flow is present. When the viscosity of the fluid strongly depends on temperature, the top–bottom symmetry is broken. In addition to the asymmetric temperature profile across the convection cell, the velocity profiles near the plate boundaries show dramatic difference from the symmetric case. We report here that the second derivative of the velocity profiles are of opposite signs in the thermal sublayers, through measurements derived from the power spectrum of temperature time-series. As a result, the stress rate applied at the plates is maintained constant within a factor of 3, while the viscosity changes by a factor of 53, in qualitative agreement with previous theory. © *1998 American Institute of Physics.* [S1070-6631(98)00306-7]

In the hard turbulence regime of thermal convection, a large scale flow, a coherent flow, is one of the main observations.[1–5] The primary thermal objects are individual plumes, which pinch off from top and bottom plates to move into the central region in a seemingly stochastic fashion. In the presence of such a global flow, there exists a velocity profile across the convection cell. If the properties of the fluid do not change with temperature, the so-called Boussinesq approximation,[5] this profile is symmetric.[6,7]

Breaking of the top–bottom symmetry is quite generic in experimental situations. When fluids change their properties with temperature, the top and bottom structures are different. We refer to this as a non-Boussinesq effect.[1,8]

In our experiment, we use glycerol as the working fluid. There, the viscosity increases dramatically as the temperature decreases.[9] This results in an asymmetric temperature field.[1] We observe that, in a certain temperature range, the velocity profiles of the top and bottom thermal layers have different shapes, specifically the second derivatives with respect to the vertical coordinate are of opposite sign. The velocity profile measurements are indirect, but are inferred from the local temperature time series. This technique was used in earlier works.[1,3,7,10]

The experimental setup is the same as the one used in our previous work.[1] The convection cell is a cubic box made of acrylic glass, 183 mm of characteristic length. It is heated uniformly from below with a Kapton foil heater, and is cooled at the top with a temperature regulated water circulation system. The temperature stability of the bottom and the top plate is about 0.2 °C and 0.04 °C, respectively. The Rayleigh number, $\mathrm{Ra}=\alpha g \Delta T L^3/\nu\kappa$, which is the control parameter in the experiment, is above 10^8. Along the central vertical axis, we insert three thermistors[11] mounted on a moving stage, which is driven by a stepping motor with micrometer precision, and thus measure the thermal boundary layers. The time series of temperature fluctuations are recorded using a lock-in amplifier and a HP basic station. From the cutoff frequency f_c of the Fourier transform of the temperature fluctuations, a velocity value can be deduced, as shown in previous work.[7]

The cutoff frequency f_c is related to the velocity by $f_c = U/\Lambda$, where U is the local velocity and Λ represents the length scale of the thermal plume, of the order of the thermal layer thickness λ_t. The measured velocity is taken to be the component parallel to the plate. This approximation was shown to be accurate when the probe is not too far from the boundaries.

In the experiment, the top and bottom temperature are held at 10.0 °C and 93.6 °C, respectively. The central region temperature is 64.9 °C and $\mathrm{Ra}=5\times10^8$. Here, the Rayleigh

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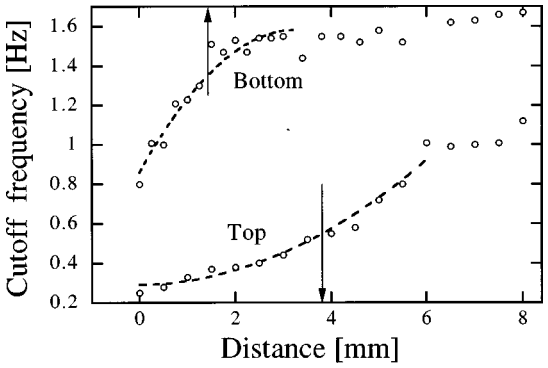


FIG. 1. Cutoff frequency profiles for the bottom and top. The two vertical arrows indicate the respective positions of the thermal boundary layers. The fitting of the curves are only near the vicinity of the thermal boundary layer.

number is calculated based on the values of glycerol diffusivities and thermal expansion coefficient for the temperature at the midpoint of the cell, which is the temperature of the majority of the fluid. The thermal boundary layer thickness is 3.8 mm for the top and 1.4 mm for the bottom. Figure 1 shows the profile of the cutoff frequency f_c as a function of the distance from the plates. Only the part near the plates is presented.[12] In both cases, the velocity increases monotonically with increasing distance from the plates ($Z=0$). But the second derivative of the profiles show opposite sign; positive for the top plate and negative for the bottom one. Figure 2 shows how $f_c\lambda_t$, which has the dimension of velocity, changes with distance. The two distances are normalized by their respective thermal boundary layer thickness.

Previously, in our work on non-Boussinesq convection, we studied the dynamic regulation of the mean central temperature by the thermal boundary layers. We presented a steady state theory based upon, among other assumptions, constancy of viscous stress in the thermal sublayers. The mean temperature obtained from this theory agreed reasonably well with the experimental measurements.[1] This theory was a crude representation of advection of heat into a plume as a steady process, from flow created by the pattern of upwellings and downwellings. However it is also reasonable to apply the theory to the *mean* flow, and then to the mean velocity in the viscous and thermal boundary layers which it sets up. This is very approximate since the temperature equation of the theory is nonlinear and the constant stress assumption is in any case a fairly crude approximation in our

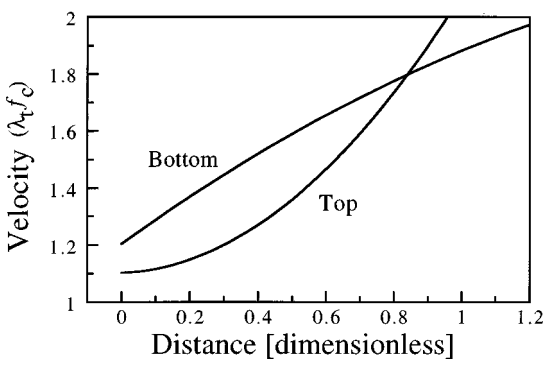


FIG. 2. Fitting plot of $\lambda_t \cdot f_c$ using data presented in Fig. 1 vs dimensionless distance, normalized by the respective thickness of the thermal layer.

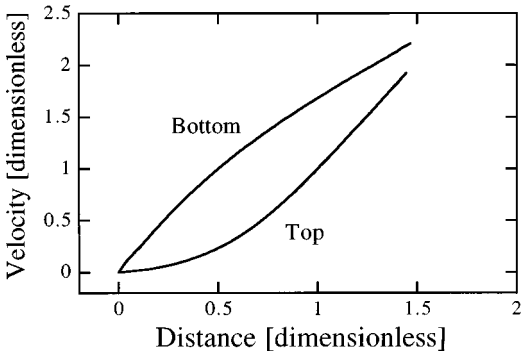


FIG. 3. Theoretical velocity profile near top and bottom plate, based on the fixed stress rate assumption. The distance is dimensionless, normalized by the respective thermal layer thickness.

parameter range; but the model should be faithful to qualitative features of the velocity field. In the example described below we use the temperature boundary-layer equation of Ref. 1 to determine the profile of velocity in the thermal layers at the same stations in the upper and lower boundary layers, so that their shapes may be compared with the measured profiles. To study the boundary layer we must identify the thermal layer and see how the velocity profile within it conforms to a constant viscous stress. This portion of the velocity profile then matches to an isothermal viscous outer layer, where the stress varies, although in practice the transition from the thermal to the outer layers is smooth.

The constant stress assumption, leads to the curves in Fig. 3. These profiles are extracted from the boundary-layer theory, using the integral equation derived in Ref. 1 [Eq. (10)]. We recall that this equation determines, in Von-Mises boundary layer coordinates, the nonlinear development of a steady thermal layer with velocity determined by a condition of fixed viscous stress. The velocity profiles are compared at the same horizontal position, and at the same stress, but appropriate wall temperatures, at the top and bottom plates. Here the top and bottom temperatures are 17.4 °C and 83.4 °C, with central temperature at 65.0 °C. The distances in the plot are normalized by the respective thermal boundary layer thickness, and velocity units are the same and normalized by the velocity value at the edge of the *top* thermal layer. The thermal layer thickness is defined here by the point where local viscosity fell within 5% of the value at the central mean temperature. Using the thermal data of glycerol, the profiles capture the peculiar behavior of the thermal layer velocity field.

From Figs. 2 and 3, we find that experiment and theory are in qualitative agreement. The constant stress assumption is essentially correct. To check the constant stress assumption in our experiment, we use the second order polynomial fits to the curves (Fig. 1). The closest fits have the form $f_c(Z)=0.86+0.44Z-0.067Z^2$ and $f_c(Z)=0.29+0.0032Z+0.017Z^2$, for bottom and top, respectively. Units here are hertz and millimeters. Stress rate is defined as $\tau=\eta(Z)\,[\partial U(Z)/\partial Z]$, where $\eta(Z)$ is the dynamic viscosity of the local fluid at Z. From our experimental data, we measure that the average central region temperature is 64.9 °C. The viscosity of glycerol drops from 35.1 poise at the top plate to 0.65 poise at the edge of the thermal layer (where

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$Z \sim 3.8$ mm), changing by a factor of 53. However the horizontal viscous stress changes by a factor of only 3! For the bottom, the viscosity changes much less, by a factor of 3–4 from the plate to the edge of the thermal layer (from 0.18 to 0.65 poise). The stress calculated is again changing only within a factor of 2. We see that in both cases, the velocity compensates for the changing viscosity, trying to keep the stress constant within a small factor. This behavior, given the nearly exponential decay of glycerol viscosity with temperature, is thus responsible for the peculiar shape of the velocity profiles.

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[11] Thermometrics, thermistors, type AA6A8-GC11KA143C/37C.
[12] We notice that the cutoff frequency at $Z=0$ does not vanish, as required by the nonslip boundary condition, for both plates. This is due to the intrinsic limitation of the measurement itself. The physical size of the probe itself spans about 0.3 mm and the extended metallic leads also introduce nonlocal disturbances by heat conduction. In the case when we measure temperature time series with a thermistor embedded inside the plate, we still get a nonzero cutoff frequency at the range from 0.12 to 0.30 Hz.

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